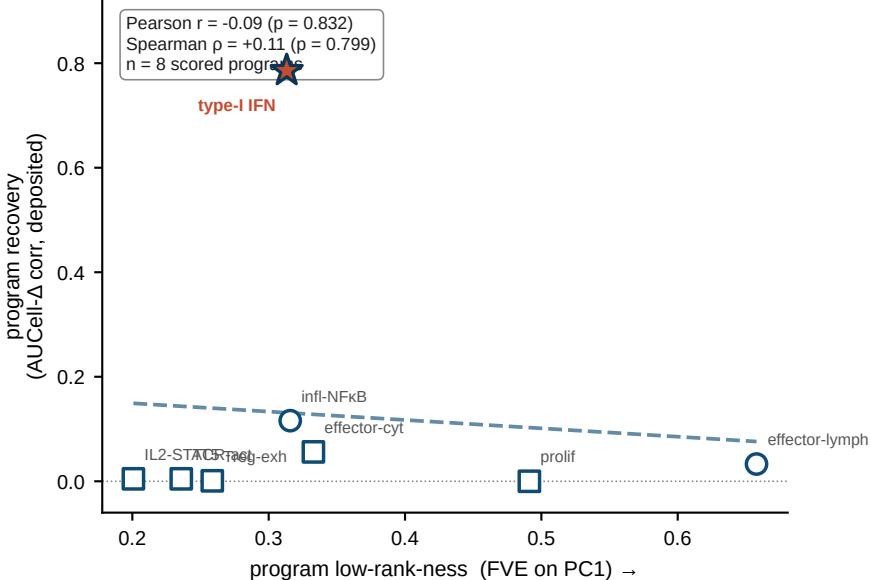


Program recovery is set by signal magnitude, not by low-dimensionality

★ type-I IFN (T5) ○ T5 program (OP3) □ T3 program (5×CRISPR)

a Recovery vs low-rank-ness (the asserted law)

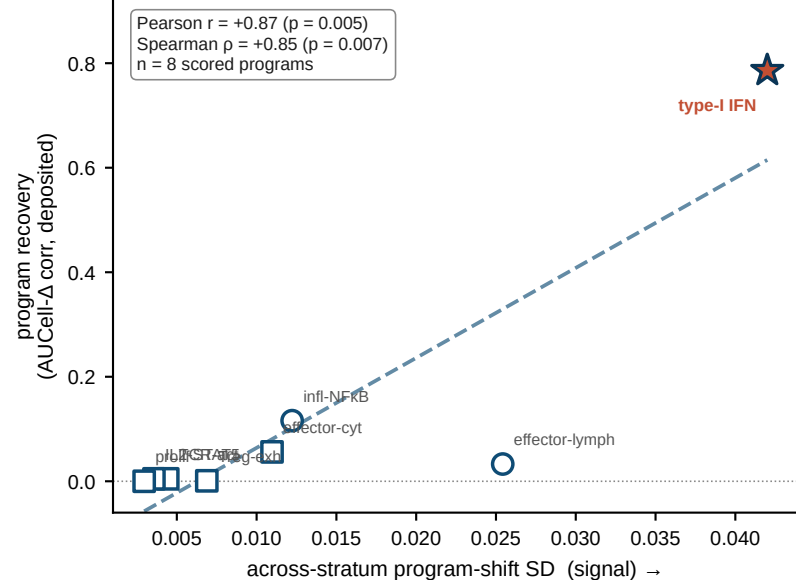
x = fraction of variance on PC1 of the program gene block



Law predicts a POSITIVE slope (more low-rank → recovered). Observed: flat / non-significant; the most low-rank program (effector-lymph, FVE-PC1=0.66) is essentially NOT recovered.

b Recovery vs program-shift signal (signal magnitude)

x = SD of observed program- Δ across perturbation strata



Recovery tracks the learnable SIGNAL, not dimensionality: type-I IFN survives because it has the largest cross-perturbation program shift, not because it is the most low-dimensional.